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The spectral kurtosis: a useful tool for characterising non-stationary signals

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Abstract

The spectral kurtosis (SK) is a statistical tool which can indicate the presence of series of transients and their locations in the frequency domain. As such, it helpfully supplements the classical power spectral density, which as is well known, completely eradicates non-stationary information. In spite of being particularly suited to many detection problems, the SK had rarely been used before now, probably because it lacked a formal definition and a well-understood estimation procedure. The aim of this paper is to partly fill these gaps. We propose a formalisation of the SK by means of the Wold–Cramér decomposition of "conditionally non-stationary" processes. This definition then engenders many useful properties enjoyed by the SK. In particular, we establish to which extent the SK is capable of detecting transients in the presence of strong additive noise by finding a closed-form relationship in terms of the noise-to-signal ratio. We finally propose a short-time Fourier-transform-based estimator of the SK which helps to link theoretical concepts with practical applications. This paper is also a prelude to a second paper where the SK is shown to find successful applications in vibration-based condition monitoring.
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1. Introduction

The spectral kurtosis (SK) was first introduced by Dwyer in [1], as a statistical tool which "can indicate not only non-Gaussian components in a signal, but also their locations in the frequency domain". Dwyer initially used it as a complement to the power spectral density, and demonstrated how it efficiently supplements the latter in problems concerned with the detection of transients in noisy signals [2,3].

Dwyer originally defined the SK as the normalised fourth-order moment of the real part of the short-time Fourier transform, and suggested using a similar definition on the imaginary part in parallel.

Since then, the SK had been seldom brought into play [5], until Pagnan and Ottonello proposed a modified definition based on the normalised fourth-order moment of the magnitude of the short-time Fourier transform [6,7]. This led to considerably simplified properties. Pagnan and Ottonello also showed that the SK could be used as a filter to recover randomly occurring signals severely corrupted by additive stationary noise.

The SK was lately given a more formal definition in [8] in light of the theory of higher-order statistics. Capdevielle defined the SK as the normalised fourth-order cumulant of the Fourier transform, i.e. as a slice of the tricoherence spectrum, and used it as a measure of distance of a process from Gaussianity. Her definition applied well to stationary signals, but encountered some difficulties with non-stationary signals. The stationary case was recently investigated in more depth by Vabrie, who proposed some interesting applications to the characterisation of harmonic processes [9,10].

There is still a need today for a correct formalisation of the SK of non-stationary processes. We believe that filling this gap is necessary for the SK to really capture the interest it deserves. Unfortunately, this task has been hindered by some theoretical difficulties:

- how can the SK—which is estimated by time averaging—detect *non-stationary* signals?
- why is the SK—which is inherently a tool for *non-Gaussian* signals—so well adapted to characterising *non-stationary* signals?
- can the correct definition of the SK of non-stationary signals be based on the assumption of circularity, which theoretically holds only for stationary signals?

In this paper, we propose a formalisation of the SK by means of the Wold–Cramér decomposition of "conditionally non-stationary" (CNS) processes. The paradigm of CNS is a natural idea we introduce here for convenience to solve the aforementioned difficulties. It basically makes it possible to establish under which conditions a "non-stationary" process (i) generates a non-Gaussian distribution, and (ii) can be described by time-averaged—i.e. stationarised—statistics. In contrast to earlier references, we will also provide an interpretation of the SK as a measure of temporal dispersion of the time–frequency energy density of a process. This point of view will shed new light on the comprehension of the SK and on some of its properties.

Overall, this paper brings together a number of fully original results in order to provide a more comprehensive view of some previously published material. Most of the content of Sections 2 (starting from the concept of conditional non-stationarity) to 5 is new—or at least generalises earlier works—unless specifically stated otherwise. All the proofs presented in the appendix are fully original.

2. Representation of non-stationary signals

2.1. The Wold–Cramér decomposition

As is well known, non-stationarity is a non-property. Therefore, there is no unique way to describe it. Since we are aiming at describing non-stationary random vibrations, we seek a stochastic representation which preserves as much as possible physical understanding.

In the stationary case, Wold's decomposition uniquely describes any stationary stochastic process as the output of a causal, linear, and time-invariant system excited by strict white noise:

$$Y(t) = \int_{-\infty}^{t} h(t-\tau)X(\tau)\,\mathrm{d}\tau. \tag{1}$$

Wold's decomposition imposes no other restriction on $X(t)$ than having a flat spectrum almost everywhere. However, for sake of simplicity, we shall also assume in the remaining of the paper that $X(t)$ has a symmetric probability density function. The frequency counterpart of Wold's decomposition is known as Cramér's decomposition (1), viz

$$Y(t) = \int_{-\infty}^{+\infty} \mathrm{e}^{j2\pi ft} H(f)\,\mathrm{d}X(f), \tag{2}$$

where the transfer function $H(f)$ is the Fourier transform of $h(s)$ (s is a dummy variable for time) and $\mathrm{d}X(f)$ is the spectral process associated with $X(t)$, i.e.

$$X(t) = \int_{-\infty}^{+\infty} \mathrm{e}^{j2\pi ft}\,\mathrm{d}X(f). \tag{3}$$

In Eq. (2), $\mathrm{e}^{j2\pi ft}H(f)\,\mathrm{d}X(f)$ may be interpreted as the result of filtering $Y(t)$ with an infinitely narrow-band filter centred on frequency f. This representation of a stationary process has the advantage of being physically meaningful. A natural solution for extending the Wold–Cramér decomposition to *non-stationary* processes is to make the filter $h(s)$ time-varying. Specifically, let us define $h(t,s)$ the causal impulse response at time t of a system excited by an impulse at time t–s, then

$$Y(t) = \int_{-\infty}^{t} h(t,t-\tau)X(\tau)\,\mathrm{d}\tau. \tag{4}$$

Such a representation has been shown to hold true for any non-stationary process and, most importantly, to be unique under mild regularity conditions of the impulse response $h(t,s)$ [11]. Here again, this decomposition is physically meaningful—$h(t,s)$ has the physical interpretation of a Green's function—and has been intensively discussed in the literature [12].

The frequency counterpart of (4) is

$$Y(t) = \int_{-\infty}^{+\infty} \mathrm{e}^{j2\pi ft} H(t,f)\,\mathrm{d}X(f), \tag{5}$$

where $H(t,f)$ is the Fourier transform of the time-varying impulse response $h(t,s)$ and $\mathrm{d}X(f)$ is the spectral process associated with $X(t)$. The Fourier decomposition (5) evidences an obvious similarity with the stationary case (2), except that a non-stationary process is now expressed as a

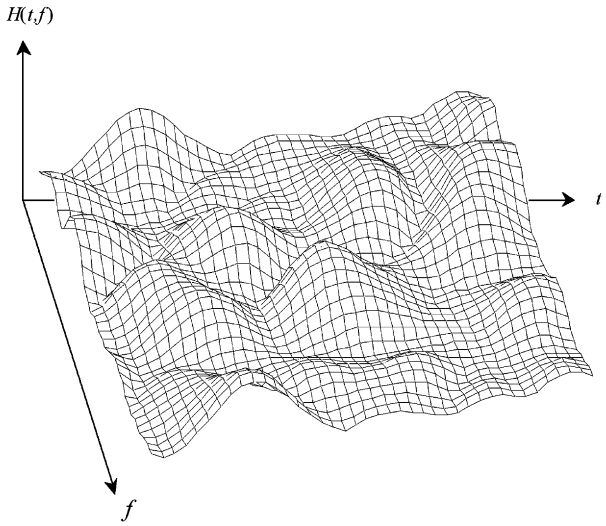

Fig. 1. Example of a complex envelope seen as the realisation of a random field.

time-varying summation of weighted complex exponentials. In Eq. (5), the time-varying transfer function $H(t,f)$ may be interpreted as the *complex envelope* or *complex demodulate* of process $Y(t)$ at frequency f, i.e. such that $e^{j2\pi ft}H(t,f)\,dX(f)$ is the output at time t of a infinitely narrow-band filter centred on frequency f.

2.2. The concept of conditional non-stationarity

Although the above Wold–Cramér decomposition of a non-stationary process involves a deterministic time-varying transfer function $H(t,f)$, there are many situations where $H(t,f)$ would be rather stochastic then deterministic, either because of random temporal variations of the filter or simply because the time datum of the process is unknown. Therefore, a more comprehensive description of $Y(t)$ is

$$Y(t) = \int_{-\infty}^{+\infty} e^{j2\pi ft}H(t,f;\varpi)\,dX(f), \tag{6}$$

where $H(t,f,\varpi)$ is a complex envelope whose shape depends on the outcome ϖ. Letting the outcome ϖ be a random variable ϖ, the complex envelope $H(t,f;\varpi)$ then becomes a random field, and the stochastic process $Y(t)$ is characterised by a double stochasticity, both in $H(t,f)$ and in $dX(f)$. This is illustrated in Fig. 1.

For simplicity, we will only consider the cases where

(i) $H(t,f;\varpi)$ is time-stationary,[1]
(ii) and is independent of the spectral process $dX(f)$.

[1]For notational simplicity we will from now on omit writing the variable ϖ so that $H(t,f)$ will stand implicitely for $H(t,f;\varpi)$ whenever there is no confusion possible.

As a consequence, such processes will be stationary in general (random outcome ϖ), but non-stationary for any particular outcome ϖ.

We shall call these processes *conditionally non-stationary* (CNS).

Note that CNS processes include stationary processes as special cases, and that any (unconditionally) non-stationary process can be made CNS by randomisation of its time datum. This fact agrees with the real-life situation where data are captured at arbitrary (random) time instants.

A typical example of a CNS process is the speech signal. It is stationary over the set of all possible sentences that can be uttered in a given time—on the average, all the signal statistics would finally end up to be time independent—but for a given sentence to be repeated many times—a realisation of $H(t,f)$—the process has time-dependent statistics and therefore is non-stationary. The next subsection presents some other typical examples of CNS processes drawn from the field of physics in general, and vibration analysis in particular.

2.3. Examples of CNS processes

2.3.1. Uniformly amplitude-modulated processes

A very simple example of a CNS process is obtained by modulating a stationary process with a stochastic envelope, i.e.

$$Y(t) = m(t;\varpi)[h(t) * X(t)]. \tag{7}$$

With respect to the decomposition of Eq. (6), this implies having $h(t,s;\varpi) = m(t;\varpi)h(s)$ and $H(t,f;\varpi) = m(t;\varpi)H(f)$. If $m(t;\varpi)$ is stationary, then the process $Y(t,\varpi)$ is itself stationary, but for any given outcome ϖ, the envelope $m(t;\varpi)$ has the effect of a deterministic amplitude modulation and consequently $Y(t)$ will be non-stationary. This is illustrated in Fig. 2.

2.3.2. Filtered-modulated white noise

This process differs from the preceding in that the stochastic modulation acts before the filter. Specifically,

$$Y(t) = h(t) * [m(t;\varpi)X(t)]. \tag{8}$$

Here, $h(t,s;\varpi) = m(t-s;\varpi)h(s)$ but $H(t,f;\varpi)$ is no longer separable in general.

2.3.3. Randomised cyclostationary processes

Cyclostationary processes are non-stationary processes whose statistics are periodically varying. Again, this is easily obtained from the proposed representation by imposing that $h(t,s)$ be a periodic function of time, i.e.

$$h(t,s) = h(t+T,s) = \sum_k h_k(s)\mathrm{e}^{j2\pi kt/T} \tag{9}$$

for a given period T. In practice, cyclostationary processes can only be observed if one keeps track of the phase reference of $h(t,s)$ [13]. If not, then the process is randomised by insertion of a random variable $t_0(\varpi)$ which accounts for the arbitrary time at which the signal is being

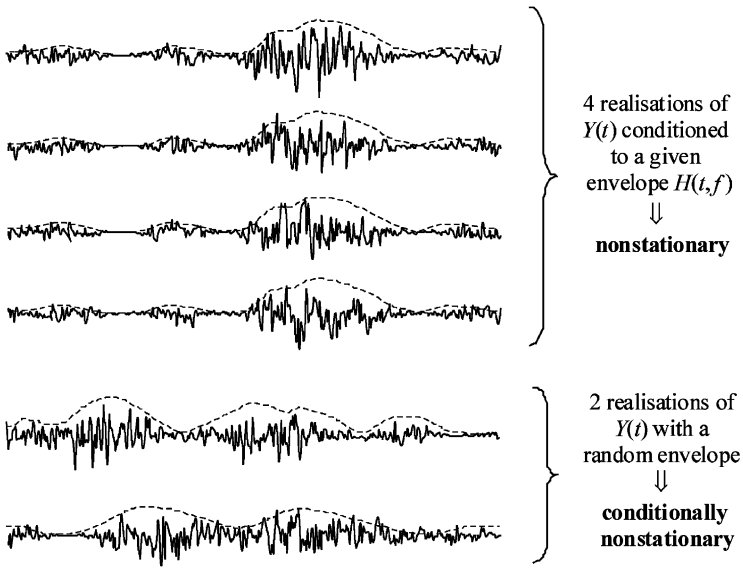

Fig. 2. Non-stationary and conditionally non-stationary visions of a uniformly amplitude-modulated process.

captured [14]

$$h(t,s) \to h(t + t_0(\varpi), s) = h(t, s; \varpi). \tag{10}$$

The randomised cyclostationary process is CNS. A typical example is provided by rotating machinery signals which are usually treated as stationary, unless they can be resynchronised with the machinery angles of rotation by means of a phase reference, in which case they are non-stationary with a periodic statistical structure.

2.3.4. Generalised shot noise

Generalised shot noise is made from a series of shaped pulses occurring at random instants. It is extremely useful for modelling numerous physical processes. A typical example is provided by rolling-element bearing signals which encompass a series of impacts in the case of incipient faults [13]—this example is fully detailed in Ref. [15].

Generalised shot noise can be obtained from setting

$$h(t, t-\tau; \varpi) = h(t-\tau) \sum_k \delta(\tau - \boldsymbol{\tau_k}(\varpi)), \tag{11}$$

where $h(t)$ is the shape of the pulses and $\boldsymbol{\tau}_k$ their random instants of occurrence. Here, the random outcome ϖ determines the values of the set $\{\boldsymbol{\tau}_k\}_{k\in\mathbb{Z}}$. After insertion in Eq. (4), this yields

$$Y(t) = \sum_k h(t - \boldsymbol{\tau}_k) X(\boldsymbol{\tau}_k), \tag{12}$$

where $X(\boldsymbol{\tau}_k)$ determines the random amplitude of the pulses. A particular case is the *generalised point process*, obtained by setting $h(t) = \delta(t)$.

Here again, the generalised shot noise process would be non-stationary if it was possible to repeat the same experiment with the first pulse occurring always at the same instant. But as soon as the first pulse is randomly distributed over the time axis the process is CNS.

2.3.5. Brownian motion

The Brownian motion starting at time t_0 is generated by setting $h(t,s) = 1$ for $0 \leqslant s \leqslant t - t_0$ and 0 elsewhere. As is well known, the Brownian motion is a non-stationary process when it is conditioned to start at a given time. If $t_0 = t_0(\varpi)$ is a random variable, then the Brownian motion is made CNS by randomisation.

2.4. Relationship between CNS and non-Gaussianity

A large class of CNS processes has the fundamental property of being characterised by non-Gaussian probability density functions. We shall prove this fundamental property in two steps:

Property 1. *Any CNS process $Y(t)$ of the form* (6) *driven by a white process $X(t)$ of order*[2] $p \geqslant 4$ *has a kurtosis greater or equal than the kurtosis of $X(t)$,*

$$\kappa_Y \geqslant \kappa_X. \tag{13}$$

Property 2. *Any CNS process $Y(t)$ of the form* (6) *driven by a white Gaussian process $X(t)$ has a non-negative kurtosis, i.e.,*

$$\kappa_Y \geqslant 0. \tag{14}$$

Property 2 says that any CNS process driven by a Gaussian process is likely to be *leptokurtic*, hence non-Gaussian. This property reveals an interesting relationship between CNS and non-Gaussianity: *Gaussian-driven CNS necessarily implies non-Gaussianity*. Based on this property, the whole reason why the spectral kurtosis—a statistical tool inherently dedicated to characterising *non-Gaussianity*—also turns out so useful for analysing *non-stationary* processes.

The idea was in essence in Ref. [3]. However, no formal justification was then given to it.

Finally, it is worth pointing out that Property 2 does not hold for Gaussian-driven *non-stationary* (as opposed to CNS) processes in general, which indeed can be shown to be Gaussian with a time-varying variance. This is why the paradigm of CNS is necessary before introducing the spectral kurtosis.

3. Statistical characterisation of the complex envelope

3.1. Spectral moments

The Wold–Cramér decomposition (6) assigns to the complex envelope $H(t,f)$ a central role for describing non-stationary processes. In the case of CNS processes, the information contained in $H(t,f)$—viewed as a random field—must be assessed by means of statistical indicators.

[2]A white process of order p is a process whose all cumulants up to order p are such that $\mathrm{Cum}\{X(t), X(t+\tau_1), \ldots, X(t+\tau_{r-1})\} = C_{rX}\delta(\tau_1)\ldots\delta(\tau_{r-1})\ \forall r \leqslant p$.

To begin with, let us consider the case where $H(t,f)$ is conditioned to a given outcome ϖ. Then, according to Section 2, the process has time-dependent statistics. Specifically, let us define the $2n$-order *instantaneous moment* $S_{2nY}(t,f)$, which measures the strength of the energy of the complex envelope at time t and frequency f:

$$S_{2nY}(t,f) \triangleq E\{|H(t,f)\,\mathrm{d}X(f)|^{2n}|\varpi\}/\mathrm{d}f = |H(t,f)|^{2n} \cdot S_{2nX} \tag{15}$$

The reason why we consider only even moments is that the stationarity of $X(t)$ implies that $\mathrm{d}X(f)$ is a *circular* process, i.e. a process whose all odd moments are nil. Interestingly enough, this has a physical meaning, since statistics on *even* moments characterise the *energy* of $H(t,f)$, e.g. $|H(t,f)|^2$. For instance, for $n = 1$, the so-defined instantaneous moment decomposes the energy contained in $Y(t)$ over the time–frequency plane (t,f):

$$S_{2Y}(t,f) = |H(t,f)|^2 \cdot \sigma_X^2 \tag{16}$$

Therefore, $S_{2Y}(t,f)$ may be interpreted as the instantaneous spectrum or the *time–frequency energy density* of signal $Y(t)$. Definition (16) ($n = 1$) is actually similar to that of *Priestley's evolutionary spectrum* [16].

Conditional statistics of the form (15) are functions of time and frequency. They are very useful for analysing the time–frequency structure of a non-stationary process, conditioned to a given outcome ϖ. But with CNS processes it is necessary to investigate how the time–frequency structure behaves on the average—i.e. by ensemble averaging on many outcomes ϖ. Such an information is conveyed by the *spectral moments*, which we shall define as

$$S_{2nY}(f) \triangleq E\{S_{2nY}(t,f)\} = E\{|H(t,f)\,\mathrm{d}X(f)|^{2n}\}/\mathrm{d}f = E\{|H(t,f)|^{2n}\} \cdot S_{2nX} \tag{17}$$

In the above equation, use has been made of the assumptions that $H(t,f)$ is (i) a time-stationary random field, (ii) is independent of $\mathrm{d}X(f)$, and (iii) that $X(t)$ is a white process of order $p \geqslant 2n$. Condition (i) means that spectral moments are functions of frequency only, while conditions (ii) and (iii) entail that $S_{2nY}(t,f)$ essentially resumes the information contained in the complex envelope.

Note that for $2n = 2$, the spectral moment

$$S_{2Y}(f) = E\{|H(t,f)|^2\} \cdot \sigma_X^2 \tag{18}$$

gives the classical *power spectral density* of $Y(t)$.

Spectral moments are very valuable statistical indicators for characterising CNS processes, but unfortunately they are not available in practice unless the experiment can be repeated an infinite number of times (i.e. to perform an ensemble average). One alternative solution is to average the instantaneous moments $S_{2nY}(t,f)$ along the time axis. Specifically, let us define the $2n$-order time-averaged moment as:

$$\langle S_{2nY}(t,f)\rangle_t \triangleq \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{+T/2} S_{2nY}(t,f)\,\mathrm{d}t, \tag{19}$$

where $\langle\cdots\rangle_t$ denotes the time-averaged operator. Then it is easy to verify that, under conditions of stationarity and ergodicity of the complex envelope $H(t,f)$,

$$S_{2nY}(f) = \langle S_{2nY}(t,f)\rangle_t, \tag{20}$$

which means that time-averaged moments, although computed for a given outcome of ϖ, are deterministic quantities independent of ϖ and identical to the spectral moments (17). In essence, the idea is similar to that first published by Welch in Ref. [18]—Welch was concerned with the computation of the power spectral density—although it applies here to CNS processes.

3.2. The spectral kurtosis

3.2.1. Definition

The previous subsection has introduced the definition of spectral moments and their expression in terms of time-averaged quantities. From these definitions, a large variety of statistical indicators can now be designed. Of particular interest for characterising CNS processes, which we have shown are likely to be non-Gaussian, are the spectral cumulants—i.e. combinations of several moments of different orders. Indeed, spectral cumulants of order $2n \geqslant 4$ have the interesting property of being non-zero for non-Gaussian processes.

The fourth-order spectral cumulant of a CNS process is defined as:[3]

$$C_{4Y}(f) = S_{4Y}(f) - 2S_{2Y}^2(f), \quad f \neq 0. \tag{21}$$

It can be shown that the larger the deviation of a process from Gaussianity, the larger its fourth-order cumulant. Therefore, the energy-normalised fourth-order spectral cumulant will give a measure of the peakiness of the probability density function of the process at frequency f.

This defines the so-called SK

$$K_Y(f) \triangleq \frac{C_{4Y}(f)}{S_{2Y}^2(f)} = \frac{S_{4Y}(f)}{S_{2Y}^2(f)} - 2, \quad f \neq 0. \tag{22}$$

3.2.2. Physical interpretation

The previous subsection has introduced the SK from a statistical point of view. Another interpretation of the SK is in terms of energy.

Recall that, according to Eqs. (18) and (20), the second-order spectral moment $S_{2Y}(f)$ can be seen as measuring the *time-average* of $|H(t,f)|^2$ at each frequency f, and so yields the power spectral density. Similarly, it may also be informative to measure the average *time-dispersion* of $|H(t,f)|^2$ at each frequency f, i.e. to measure how much the energy of the complex envelope is fluctuating in the time direction—see Fig. 3. This is given by the time-variance:

$$\langle S_{4Y}(t,f)\rangle_t - \langle S_{2Y}(t,f)\rangle_t^2 \tag{23}$$

whose scale-normalised version is

$$\frac{\langle S_{4Y}(t,f)\rangle_t}{\langle S_{2Y}(t,f)\rangle_t^2} - 1 = \frac{S_{4Y}(f)}{S_{2Y}^2(f)} - 1 = K_Y(f) + 1. \tag{24}$$

[3]The factor 2—rather than 3 as in the usual definition of cumulants—comes from the fact that $\mathrm{d}X(f)$ is a *circular* random variable. This results from the process being modelled as CNS. Factor 3 should be substituted at $f = 0$, where the incremental process $\mathrm{d}X(f)$ is real. However, this case will not be considered in the following, for it does not present actual interest.

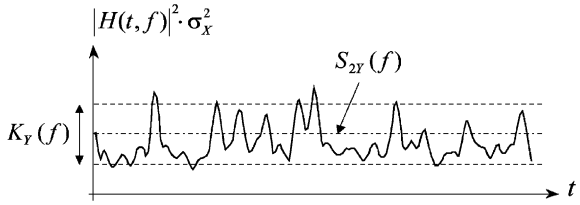

Fig. 3. Interpretation of the power spectrum and of the spectral kurtosis as the time-average and the time-dispersion of $|H(t,f)|^2$, respectively.

Up to a constant term (which actually does not contain any information), the above quantity is exactly equal to the SK as defined in Eq. (22).

We believe that the interpretation of the SK as a measure of temporal dispersion of the time–frequency energy distribution[4] is physically more meaningful than its formal definition (22) in terms of cumulants. As a matter of fact, if one remembers that the Wold–Cramér decomposition (5) can be interpreted as a filter bank decomposition, i.e. as the summation of a series of filtered versions of the process by infinitely narrow frequency-bands, then the SK at a given frequency f is equivalent to measuring the peakedness of the squared envelope $|H(t,f)\,\mathrm{d}X(f)|^2$. As such, it is expected to be very sensitive to non-stationary patterns (transients) in a signal and to indicate exactly at which frequencies those patterns occur.

4. Properties of the SK

The previous section has shown how to characterise CNS processes by means of suitably designed statistical indicators. Of particular interest among these indicators is the SK because of its numerous properties, the most important of which we now present. Many of these properties are being introduced here for the first time. It must be understood that they are specific to the proposed definition of CNS processes in terms of the Wold–Cramér decomposition.

4.1. General properties

Property 3. *The SK of a CNS process $Y(t)$ of the form* (6) *is given by*

$$K_Y(f) = \gamma_{4H}(f)[2+\kappa_X] - 2 \geqslant \kappa_X, \quad f \neq 0, \tag{25}$$

where $\gamma_{4H}(f) = E\{|H(t,f)|^4\}/E\{|H(t,f)|^2\}^2$ and κ_X is the kurtosis of $X(t)$.

As a consequence, the following properties (4–7) hold true:

Property 4. *The SK of a CNS process $Y(t)$ driven by a Gaussian process is given by*

$$K_Y(f) = 2[\gamma_{4H}(f) - 1] \geqslant 0, \quad f \neq 0. \tag{26}$$

[4]i.e. as a normalised second-order cumulant of an energy quantity instead of a normalised fourth-order cumulant.

The inequality in Property 4 is to be compared with Property 2 previously established in Section 2. It proves that the probability density function of a Gaussian-driven CNS process is leptokurtic at any frequency. In other words, it evidences once again that the intrinsic non-stationarity of a CNS process induces a non-Gaussian distribution. This is in plain contrast with stationary processes, for which the following properties hold true:

Property 5. *The SK of a purely* stationary process $Y(t)$*—i.e. not* CNS*—is independent of frequency and is given by*

$$K_Y(f) = \kappa_X, \quad f \neq 0. \tag{27}$$

This property relates the SK of a stationary process to its classical kurtosis.

Property 6. *The SK of a purely* stationary Gaussian *process* $Y(t)$ *is zero*

$$K_Y(f) = 0, \quad f \neq 0. \tag{28}$$

This property was earlier recognised by Dwyer in [1] and used for designing a detection test of transients in additive Gaussian noise.

4.2. The SK of some typical signals

Property 7. *The SK of the uniformly amplitude-modulated process* (7) *is given by*

$$K_Y(f) = \gamma_{4m}(\kappa_X + 2) - 2, \quad f \neq 0, \tag{29}$$

where $\gamma_{4m} = E\{|m(t)|^4\}/E\{|m(t)|^2\}^2$.

Property 8. *The SK of a modulated tone* $Y(t) = \boldsymbol{A}(t)\exp(j2\pi f_0 + j\boldsymbol{\phi})$, *where* $\boldsymbol{A}(t)$ *is a stationary complex envelope and* $\boldsymbol{\phi}$ *a random phase, is given at* $f = f_0$ *by*

$$K_Y(f_0) = \gamma_{4A} - 2, \quad f = f_0, \tag{30}$$

where $\gamma_{4A} = E\{|\boldsymbol{A}(t)|^4\}/E\{|\boldsymbol{A}(t)|^2\}^2$ (*note that in the present case the SK is not defined at* $f \neq f_0$). *This is obtained as a special case of Eq.* (6), *where the orthogonal process* $\mathrm{d}X(f)$ *is a random impulse. If* $\boldsymbol{A}(t)$ *is a deterministic constant, then* $K_Y(f_0) = -1$ *which coincides with the result proved in* [8,9].

Particular cases of Property 8 were first proved in Refs. [6,7] and then reintroduced in Refs. [8–10]. In [10] the authors present interesting applications of this property to the characterisation of harmonics.

4.3. The SK of a signal in additive noise

Property 9. *The SK of a CNS process $Z(t) = Y(t) + N(t)$, where $N(t)$ is an additive stationary noise independent of $Y(t)$, is given by*

$$K_Z(f) = \frac{K_Y(f)}{[1+\rho(f)]^2} + \frac{\rho(f)^2 K_N}{[1+\rho(f)]^2}, \quad f \neq 0, \tag{31}$$

where $\rho(f) = S_{2N}(f)/S_{2Y}(f)$ is the noise-to-signal ratio.

Property 10. *The SK of a CNS process $Z(t) = Y(t) + N(t)$, where $N(t)$ is an additive stationary Gaussian noise independent of $Y(t)$, is given by*

$$K_Z(f) = \frac{K_Y(f)}{[1+\rho(f)]^2}, \quad f \neq 0. \tag{32}$$

A similar property was mentioned in Ref. [7], for the specific case where $Y(t)$ is a linear combination of pure tones. More generally, it is worth insisting on the important potential of Property 10 in detection problems. Indeed, there are many situations where the signal to detect has a known SK—e.g. of the form (29) or (30)—and is embedded in stationary Gaussian noise of unknown colour. Then Eq. (32) offers the rare opportunity to blindly estimate the noise-to-signal ratio $\rho(f)$, from which a large variety of detection filters can then be designed. For example, the *Wiener filter* $W(f)$ is the filter that best extracts the signal $Y(t)$ from the noisy measurement $Z(t)$ and is expressed as

$$W(f) \triangleq \frac{1}{1+\rho(f)}. \tag{33}$$

Alternatively, the *matched filter* is the filter that maximises the signal-to-noise ratio of the recovered signal; it is given by the eigenvector associated with the largest eigenvalue of the autocorrelation matrix which has for elements the inverse Fourier transform of

$$M(f) \triangleq \frac{1}{\rho(f)}. \tag{34}$$

In all cases, the required detection filter only depends on the unknown noise-to-signal ratio $\rho(f)$ which can be estimated from the relationship

$$[1+\rho(f)]^2 = \frac{K_Y(f)}{K_Z(f)}, \quad f \neq 0. \tag{35}$$

According to the author's knowledge, the connection between the SK, the Wiener filter, and the matched filter is an original finding, although it bears similarities with some recent work in signal processing [17]. The idea was suggested in Refs. [6,7] where the authors used the raw SK as a denoising filter. However, their practice had been given no theoretical justification and Eq. (33) indicates that the *square root* of the SK rather than the SK itself is the optimal denoising filter. Applications of the SK to Wiener filtering and matched filtering is further discussed in Ref. [15].

5. Estimation issues

Up to here, our aim has been to provide the SK with a theoretical framework from which sound definitions and properties could be derived. Actually, this framework is also helpful for designing an estimator of the SK, a necessary step for connecting theoretical results with real life practice.

As is always the case with estimation issues, there are several plausible candidates for an estimator. Our concern here is to propose a simple one, which shares a close connection with the Wold–Cramér decomposition of CNS processes. Such an estimator can be built from the short-time Fourier transform (STFT).

5.1. STFT decomposition of a CNS process

Let $Y(n)$ be the sampled version of process $Y(t)$ where it is assumed for simplicity and without loss of generality that the sampling period is equal to 1. Then, for a given (positive) analysis window $w(n)$ of length N_w and a given temporal step P, the STFT of process $Y(n)$ is defined as

$$Y_w(kP,f) \triangleq \sum_{n=-\infty}^{\infty} Y(n)w(n-kP)\mathrm{e}^{-j2\pi nf}. \tag{36}$$

Furthermore, if the analysis window $w(n)$ fulfils some mild conditions [19], then the process $Y(n)$ has the STFT representation

$$Y(n) = \int_{-1/2}^{+1/2} Y_w(n,f)\mathrm{e}^{j2\pi nf}\,\mathrm{d}f. \tag{37}$$

Eq. (37) may be viewed as the discrete counterpart of Eq. (5). However, the equality of the integrals does not imply the equality of their integrands.

Proposition 1. *For the* STFT $Y_w(kP,f)$ *to be identified with the integrand of Eq.* (5), *it is necessary that the following conditions hold*:

C1: $H(n,f)$ *has slow temporal variations in n as compared to the window length* N_w,
C2: $H(n,f)$ *has slow frequency-variations in f as compared to the spectral bandwidth of* $w(n)$.

Condition C2 is common to any spectral estimator, whereas condition C1 imposes that the analysis window covers intervals over which the signal is quasi-stationary, or in other words that the analysis window samples the complex envelope sufficiently fast so that no information is lost in terms of Shannon's sampling theorem.

By denoting τ_H^t and τ_H^s the correlation lengths of $h(t,s)$ with respect to time t and time-lag s, respectively, conditions C1 and C2 are more concisely expressed as

$$\tau_H^s \ll N_w \ll \tau_H^t \tag{38}$$

i.e. *the window length* N_w *should be longer than the signal correlation length and shorter than the temporal variation of its spectral content*. As a matter of fact, this implies that the STFT-based estimator applies only to processes whose time-varying impulse response $h(t,s)$ has a slow evolution in time t as compared to its effective correlation time in s (Fig. 4).

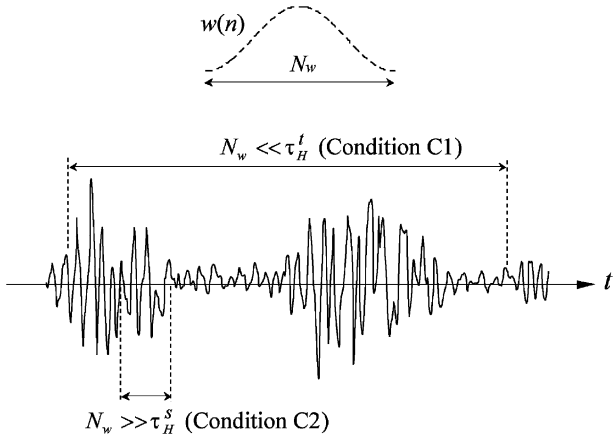

Fig. 4. Conditions on the analysis window for the STFT analysis of an oscillatory process.

Conditions C1 and C2 actually turn out to be common to many time–frequency estimators. Processes satisfying these conditions have been coined *oscillatory* by Priestley [16]. They can also be related to the class of *locally stationary processes*, as introduced by Silverman [21].

5.2. STFT-based estimation of the spectral kurtosis

5.2.1. Definition

First, let us define the $2n$th order *empirical* spectral moment of $Y_w(kP,f)$ as

$$\hat{S}_{2nY}(f) \triangleq \langle |Y_w(kP,f)|^{2n} \rangle_k \tag{39}$$

with $\langle \cdots \rangle_k$ standing for the time-averaged operator over index k. For instance, for $n = 1$, $\hat{S}_{2Y}(f)$ is an *estimator* of the power spectral density of $Y(n)$—e.g. as classically done in Welch's method [18]. It should be pointed out that, strictly speaking, the so-defined spectral moments are functions of the analysis window $w(n)$ and of the temporal step P. However, for simplicity, we shall drop these dependences on w and P in the notations.

Then, the STFT-based estimator of the SK can be defined as:

$$\hat{K}_Y(f) \triangleq \frac{\hat{S}_{4Y}(f)}{\hat{S}_{2Y}^2(f)} - 2, \quad |f - \mathrm{mod}(1/2)| > \frac{1}{N_w}. \tag{40}$$

The proposed estimator (40) shares close similarities with the historical definition of the SK as first introduced in [1–3]. Note that it is also very similar to the proposed estimator of [6–9]. But in contrast to these references, our estimator has been explicitly deduced from a time–frequency approach.

5.2.2. Interpretation

In terms of the STFT, $Y_w(kP,f)$ is the complex demodulate obtained by narrowband filtering signal $Y(n)$ around frequency f [19]. Hence, the STFT-based SK is to be interpreted as measuring

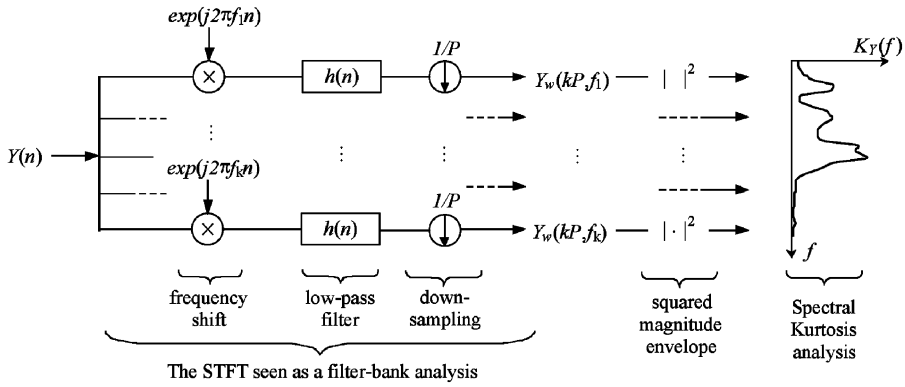

Fig. 5. Interpretation of the STFT-based SK as a measure of time-dispersion of the energy of the envelope at the output of a filterbank.

the temporal dispersion of the energy of the envelope $Y_w(kP,f)$. This is exactly the same interpretation as suggested in Section 3.2 for the SK in terms of the Wold–Cramér decomposition. This interpretation, in conjunction with a filter-bank vision of the STFT, is illustrated in Fig. 5.

5.2.3. *Statistical performance of the STFT-based SK*

5.2.3.1. Bias. The analysis of the bias of the STFT-based SK is not easy. There are basically two sources of bias, stemming from (i) finite sample effects and (ii) leakage effects due to the analysis window $w(n)$.

Finite sample effects originate from the fact that, in practice, the summation $\langle \cdots \rangle_k$ cannot be done on an infinite amount of data. This source of bias has been investigated in [9] (for stationary signals) where the authors have proposed a compensated estimator based on the use of k-statistics.

The second source of bias is inherent to the STFT and unfortunately cannot be compensated for. It is particularly critical here, where non-stationary signals must be analysed with short windows $w(n)$. One notable manifestation of the leakage is in the vicinity of $f = 0 \bmod (1/2)$, where the theoretical discontinuity in the SK spreads over a bandwidth of the order $1/N_w$ instead of being concentrated at one single point. This is why the condition $f \neq 0 \bmod (1/2)$ in definition (40) has been replaced by the safer condition $|f - \bmod (1/2)| > 1/N_w$.

When finite sample effects can be neglected—which is a reasonable assumption as compared to the importance of leakage effects—then large sample results can be used to find an approximate bias of $\hat{K}_Y(f)$.

Proposition 2. *Under conditions* C1 *and* C2, *the STFT-based* SK (40) *of a CNS process has the approximate and asymptotic bias*:

$$E\{\hat{K}_Y(f)\} - K_Y(f) \approx \gamma_{4H}(f) \cdot \kappa_X \left(\frac{\gamma_{4w}}{N_w} - 1 \right) \underset{N_w}{\rightarrow} -\gamma_{4H}(f) \cdot \kappa_X, \quad |f - \mathrm{mod}(1/2)| > \frac{1}{N_w}, \tag{41}$$

where $\gamma_{4w} = N_w|\sum_n |w(n)|^4|/|\sum_n |w(n)|^2|^2$ *is the time-bandwidth product of the square of the analysis window.*

Proposition 1 establishes that the STFT-based SK is generally biased, except when the CNS process under analysis is Gaussian driven, i.e. when $\kappa_X = 0$—this result is to be related to the classical kurtosis which is generally biased except in the Gaussian case. In all other situations, the bias of the STFT-based SK is proportional to $(\gamma_{4w}/N_w - 1)$. For example, with a rectangular and a Hanning window, $\gamma_{4w} = 1$ and $\gamma_{4w} = \frac{35}{18} \approx 2$, respectively. In general, the larger the bandwidth of the analysis window, the greater the induced bias. Note that $(\gamma_{4w}/N_w - 1)$ rapidly decreases to -1 as N_w becomes large. Hence, *the STFT-based SK of a non-Gaussian-driven process tends as* $O(1/N_w)$ *to the SK of the equivalent Gaussian-driven process.*

5.2.3.2. Variance. For simplicity, we consider only the case where $Y(t)$ is a Gaussian-driven CNS process.

Proposition 3. *Under conditions* C1 *and* C2, *the STFT-based SK of a Gaussian-driven CNS process has the approximate and asymptotic variance*:

$$\mathrm{Var}\{\hat{K}_Y(f)\} \approx 2\,\frac{K_Y(f)+2}{K}\left(3\,\frac{S_{8H}(f)}{S_{4H}^2(f)} + 4\,\frac{S_{4H}(f)}{S_{2H}^2(f)} - \frac{6S_{6H}(f)}{S_{4H}(f)S_{2H}(f)}\right), |f - \mathrm{mod}(1/2)| > \frac{1}{N_w}, \tag{42}$$

where K is the number of time averages used in the estimate and $S_{2nH}(f) \triangleq E\{|H(t,f)|^{2n}\}$.

Proposition 2 establishes that the STFT-based SK has a variance proportional to $K_y(f)+2$, and inversely proportional to the number of averages K. When the complex envelope $H(t,f)$ is Gaussian distributed, then result (42) drastically simplifies as

$$\mathrm{Var}\{\hat{K}_Y(f)\} \approx 16\,\frac{K_Y(f)+2}{K}, \quad |f - \mathrm{mod}(1/2)| > \frac{1}{N_w}. \tag{43}$$

Note also that in the stationary case—obtained by setting $H(t,f) = H(f)$ deterministic in (42)—the STFT-based SK has variance

$$\mathrm{Var}\{\hat{K}_Y(f)\} \approx 2\,\frac{K_Y(f)+2}{K}, \quad |f - \mathrm{mod}(1/2)| > \frac{1}{N_w}, \tag{44}$$

which agrees with the result derived in Ref. [9] for stationary Gaussian signals.

It is also worth mentioning that the proposed STFT-based SK has a much lower variance than the estimator initially proposed by Dwyer in [1].

5.2.4. Setting the temporal step P

In principle condition C1 ensures that P can be set equal to N_w in Eq. (36). However, care should be taken when computing the kurtosis of $Y_w(kP,f)$ in order to obtain shift-invariant results, i.e. that are independent of the time datum t_0 of the grid $\{t_0, t_0 \pm P, t_0 \pm 2P, t_0 \pm 3P, \ldots\}$ on which the STFT $Y_w(kP,f)$ is sampled. The issue is akin to ensuring that the second power of the complex envelope $Y_w(kP,f)$ meets the Shannon condition, i.e. is not under-sampled. Since the bandwidth of the complex envelope $Y_w(kP,f)$ is on the order of $1/N_w$, that of its second power is

on the order of $2/N_w$. Hence the Shannon condition yields:

$$P \leqslant \frac{N_w}{4}. \tag{45}$$

Therefore at least 75% overlap should be used with the proposed STFT-based SK.

5.2.5. Setting the window length N_w

The basic idea behind the SK is to get a quantity that ideally takes high values when the signal contains transients and is zero when the signal is stationary Gaussian. The proposed STFT-based SK enjoys these properties provided the length of the analysis window fulfils conditions C1 and C2. Whereas condition C2 is typical to any spectral estimator—and applies equally well to stationary and non-stationary processes—condition C1 is very specific to the class of spectral estimators introduced in Section 3 for CNS processes.

A similar requirement to condition C1 was mentioned in Reference [3], but from a different point of view. The argument is that the SK can only detect transients if the temporal duration of the analysis window $w(n)$ is short enough, otherwise the STFT would tend to Gaussianity by the central limit theorem and thus the SK would tend to zero [22].

This can actually be verified as follows. For an integration time $N_w > \tau_H^t$ (violation of condition C1) the function $H(kP,f)$ in the STFT is seen as almost constant so that $\gamma_{4H}(f)$ becomes a function of N_w which tends to 1. Thus, from Proposition 1:

$$E\{\hat{K}_Y(f)\} = \gamma_{4H}(f;N_w)\left(2 + \kappa_X \frac{\gamma_{4W}}{N_w}\right) - 2 \underset{N_w}{\rightarrow} 0. \tag{46}$$

Hence, *the proposed STFT-based SK must be set with N_w as short as possible—as far as permitted by condition* C2. This makes a fundamental difference between the SK as defined in the present paper and the spectral kurtosis as defined in References [8–10] on stationary signals. In particular, our definition of the SK *is not* a slice of the tricoherence spectrum[5], in which the length of the analysis window is stretched to infinity [23].

Another difficulty concerning the choice of N_w was discussed in Ref. [7] in the case of randomly occurring impulses. Such transients are by definition very brief in time and so condition C1 cannot be fulfilled. As a consequence, the SK can be shown to have values depending on N_w. More precisely,

[5]The tricoherence spectrum of a real stationary process $Y(n)$ is defined as

$$T_{4Y}(f_1,f_2,f_3) = \lim_{N\to\infty} \frac{N^p \cdot E\{Y_N(f_1)Y_N(f_2)Y_N(f_3)Y_N^*(f_1+f_2+f_3)\}}{\sqrt{E\{|Y_N(f_1)|^2\}E\{|Y_N(f_2)|^2\}E\{|Y_N(f_3)|^2\}E\{|Y_N(f_1+f_2+f_3)|^2\}}},$$

where $Y_N(f)$ is the Fourier transform taken over N samples and $p = 0$ or 1 whether the spectrum is normalised w.r.t. power or amplitude, respectively. The evaluation of the tricoherence spectrum on the slice $f_1 = -f_2 = f_3$ yields a formula similar—but *not equal*—to Eq. (22).

Property 11. *The STFT-based SK of a point process $Y(t)$, with p the probability of occurrence of the impulses, is given by*

$$\hat{K}_Y(f) = \frac{\gamma_{4w}}{pN_w} - 2, \quad pN_w \ll 1, \tag{47}$$

where γ_{4w} is as defined in Proposition 2.

A similar result was proved in Ref. [7] in the special case where the analysis window is rectangular with no overlap (i.e. when $\gamma_{4w} = 1$ and $P = N_w$).

As a conclusion, a recognised difficulty with the STFT-based SK is that there is no rule for setting a priori the appropriate duration N_w of the analysis window $w(n)$ so that conditions C1 and C2 are automatically fulfilled. Property 11 has even exemplified a class of processes for which condition C1 cannot be met. Further research should focus on this issue, in order to make the estimator of the SK more robust with respect to the settings of its parameters.

However, from our experience, a still wise strategy is to compute the STFT-based SK for different durations N_w and then to select the value that maximises the overall level of the SK in the frequency band of interest. This technique is investigated in detail in Ref. [15], and leads to the concept of the "kurtogram".

5.3. Example

This subsection illustrates the scope of validity of the proposed STFT-based SK on a synthetic signal. The tested signal $Z(n)$ is made up of a combination of three terms:

(i) a random-phased sinusoid of frequency $f_0 = \frac{1}{16}$, amplitude modulated by another random-phased sinusoid of frequency 1/900:

$$Y_1(n) = A(n)\sin(2\pi n/16 + \phi_1) \quad \text{with } A(n) = \sin(2\pi n/900 + \phi_2), \tag{48}$$

(ii) a narrow-band random noise centred on frequency 0.3, amplitude modulated by a positive sinusoid of frequency 1/900:

$$Y_2(n) = m(n)N_2(n) \quad \text{with } m(n) = 1 + \sin(2\pi n/900)$$

and

$$N_2(n) = 1.9\cos(0.6\pi)N_2(n-1) - 0.9025N_2(n-2) + X(n) \tag{49}$$

with $X(n)$ a stationary Gaussian noise with unit variance,

(iii) a stationary Gaussian noise $N(n)$ of variance σ_N^2.

According to Properties 7, 8 and 10, the compounded signal $Z(n)$ has a theoretical SK given in the interval $0 < f < 1/2$ by:

$$K_Z(f) = \begin{cases} \dfrac{\gamma_{4m}(\kappa_X + 2) - 2}{[1 + \rho_2(f)]^2}, & f > \dfrac{1}{N_w}, \quad \dfrac{1}{2} - f > \dfrac{1}{N_w}, \quad |f - f_0| > \dfrac{1}{N_w}, \\ \dfrac{\gamma_{4A} - 2}{[1 + \rho_1]^2}, & f = f_0 \end{cases} \tag{50}$$

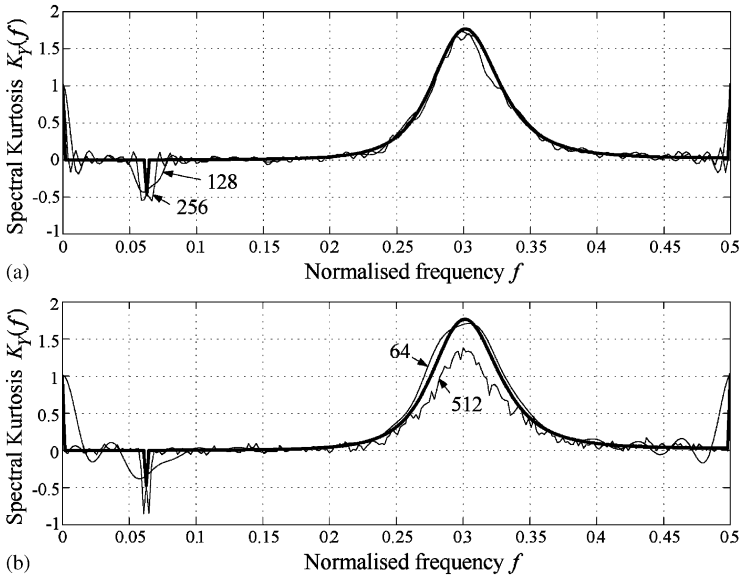

Fig. 6. STFT-based estimate of the SK versus the theoretical SK (thick line), for different window lengths N_w. Reasonable results are obtained for $128 \leqslant N_w \leqslant 256$.

with $\gamma_{4m} = \frac{35}{18}$, $\gamma_{4A} = 1.5$, $\kappa_X = 0$. The noise-to-signal ratios ρ_1 and $\rho_2(f)$ associated with signals $Y_1(n)$ and $Y_2(n)$ are given by

$$\rho_1 = \frac{4 \cdot \sigma_N^2}{\langle |A(n)|^2 \rangle_n} \frac{\varepsilon_w}{N_w} \tag{51}$$

with $\varepsilon_w = N_w \sum_n |w(n)|^2 / |\sum_n w(n)|^2$ the time-bandwidth product of the analysis window, and

$$\rho_2(f) = \sigma_N^2 |1 - 1.9\cos(0.6\pi)\mathrm{e}^{-j2\pi f} + 0.9025\mathrm{e}^{-j4\pi f}|^2. \tag{52}$$

The STFT-based SK of signal $Z(n)$ was computed using 10^6 samples and a Hanning window ($\varepsilon_w = 1.5$) with 75% overlap. Different lengths N_w were tried for the analysis window. The results are displayed in Fig. 6a, together with the theoretical SK obtained from Eq. (50). Inspection of Fig. 6a shows that reasonable estimations could be obtained for window lengths in the range 128–256. These estimates clearly illustrate the utility of the SK to detect and to characterise different non-stationary structures hidden in a noisy signal.

For window lengths shorter than 128, excessive bias was observed due to violation of condition C2. On the other hand, for window lengths greater than 256 condition C1 could not be met. These two extreme cases are illustrated in Fig. 6b, where clearly too short an analysis window ($N_w = 64$) induces leakage effects ($f = 0$, $f = f_0$, and $f = 1/2$), whereas too long an analysis window ($N_w = 512$) drags the estimated SK towards zero.

This latter point was further investigated by computing the STFT-based SK of the synthetic signal $Y(n) = m(n)X(n)$ (with $m(n)$ and $X(n)$ defined as above) for different window lengths. This white signal theoretically produces a constant SK $K_Y(f) = \gamma_{4m}(\kappa_X + 2) - 2 = 34/18$. Fig. 7

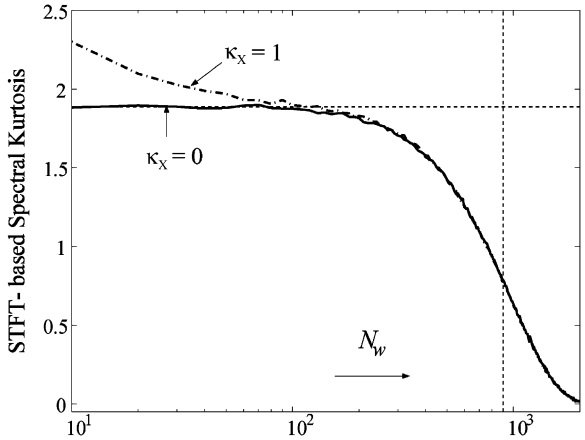

Fig. 7. Evolution of the STFT-based estimate of the SK (continuous curve) with respect to the window length N_w. The theoretical SK is 34/18 (horizontal dotted line). The vertical dotted line shows the correlation length τ_H^t above which Condition C1 is violated. The dotted curve shows the estimated SK when the underlying process is Gamma-distributed, and is to be compared with a theoretical SK value of $23/6 \approx 4$.

shows how the estimate differs from this theoretical value as N_w approaches the correlation length τ_H^t of the complex envelope. Also shown on the same figure are the results obtained when $X(n)$ is Gamma-distributed instead of being Gaussian, i.e. when $\kappa_X = 1$. This should yield a theoretically constant SK of $23/6 \approx 4$. As predicted by Proposition 1, the STFT-based SK is then severely biased, due to the non-Gaussianity of the underlying process.

6. Conclusion

The SK was heuristically introduced 20 years ago as the normalised fourth-order moment of the real part the STFT. This empirical definition has only recently been refined and formalised in the case of stationary signals. In this paper, we have proposed a parallel formalisation for non-stationary signals, by means of the Wold–Cramér decomposition and of the paradigm of conditionally non-stationary processes. CNS processes have the fundamental property of translating their intrinsic non-stationarity into non-Gaussian characteristics. Hence, they allow the definition of spectral moments and cumulants, which are non-zero in general. The SK then happens to be the normalised fourth-order spectral cumulant of a CNS process.

As originally proposed by Dwyer, the SK is expected to provide additional information about the frequency contents of transients which the traditional power spectral density cannot display. From our approach based on spectral moments, it is very clear that one supplements the other: the power spectral density is to be interpreted as a measure of position (time-average), whereas the spectral kurtosis as a measure of dispersion (time-variance) of a time–frequency energy density.

The so-defined SK enjoys many properties, many of which we have listed here for the first time. Previous works have reported that the SK is also able to detect transients in the presence of a strong background stationary noise. We have exactly established to which extent this is possible by finding a closed-form relationship between the SK and the noise-to-signal ratio. The same

result allows interesting relationships between the SK, the Wiener filter, and the matched filter, to be advantageously exploited in signal detection schemes.

We have finally proposed a STFT-based estimator of the SK which should help linking theoretical concepts with practical applications. Contrary to the stationary case, the STFT-based SK encounters a number of difficulties with non-stationary signals. It can only be computed for a certain class of processes, whose spectral components are slowly varying—the so-called Priestley's class of oscillatory processes. The very critical parameter in the STFT-based SK is the length of the analysis window. If set too short, it induces excessive leakage bias. On the other hand, if set too large—over a limit which depends on the process—the SK rapidly tends to zero according to the Central Limit theorem. Moreover, we have shown that the STFT-based SK is systematically biased when applied to non-Gaussian driven CNS processes, and that at least 75% overlap should be used in order to obtain shift-invariant results. However, these difficulties are somehow balanced by the fact that the STFT-based SK is the only simple estimator of the SK available to date.

Further research should focus on increasing the stability of the SK estimator. Indeed, having defined the SK as a measure of temporal dispersion of a time–frequency energy density, many variations are possible by using concurrent definitions of dispersion, of time–frequency density, or both.

Appendix of proofs

Proof of Property 1. Using the fact that $X(t)$ is a stationary white process of variance σ_X^2 such that (by definition) $E\{\mathrm{d}X(f)\,\mathrm{d}X(f2)\} = \sigma_X^2\,\mathrm{d}f_1\,\mathrm{d}f_2\delta(f_1+f_2)$, it is easy to show that the autocorrelation function of $Y(t)$ has decomposition

$$E\{Y(t_1)Y(t_2)\} = \sigma_X^2 \int_{-\infty}^{+\infty} \mathrm{e}^{j2\pi f(t_1-t_2)} E\{H(t_1,f)H^*(t_2,f)\}\,\mathrm{d}f$$

from which it follows that

$$E\{Y(t)^2\} = \sigma_X^2 E\left\{\int |H(t,f)|^2\,\mathrm{d}f\right\}.$$

Similarly, for a stationary white noise of order $p \geqslant 4$, i.e. such that (by definition) $E\{\mathrm{d}X(f_1)\,\mathrm{d}X(f_2)\,\mathrm{d}X(f_3)\,\mathrm{d}X(f_4)\} = \mathrm{d}f_1\,\mathrm{d}f_2\,\mathrm{d}f_3\,\mathrm{d}f_4[\sigma_X^2\delta(f_1+f_2)\delta(f_3+f_4) + \sigma_X^2\delta(f_1+f_3)\delta(f_2+f_4) + \sigma_X^2\delta(f_1+f_3)\delta(f_2+f_4) + C_{4X}\delta(f_1+f_2+f_3+f_4)]$ with C_{4X} the fourth-order cumulant of $X(t)$, it follows that

$$\begin{aligned} E\{Y(t)^4\} &= 3\sigma_X^2 E\left\{\left(\int |H(t,f)|^2\,\mathrm{d}f\right)^2\right\} \\ &\quad + C_{4X} E\left\{\iiint H(t,f_1)H(t,f_2)H(t,f_3)H(t,-f_1-f_2-f_3)\,\mathrm{d}f_1\,\mathrm{d}f_2\,\mathrm{d}f_3\right\} \\ &= 3\sigma_X^4 E\left\{\left(\int |h(t,s)|^2\,\mathrm{d}s\right)^2\right\} + C_{4X} E\left\{\int |h(t,s)|^4\,\mathrm{d}s\right\}. \end{aligned}$$

Therefore,

$$\kappa_Y = \frac{E\{Y(t)^4\}}{E\{Y(t)^2\}^2} - 3 = \frac{3E\{(\int |h(t,s)|^2\,\mathrm{d}s)^2\} + \frac{C_{4X}}{\sigma_X^4} E\{\int\ |h(t,s)|^4\,\mathrm{d}s\}}{(E\{\int |h(t,s)|^2\,\mathrm{d}s\})^2} - 3.$$

Finally, recognising that $C_{4X}/\sigma_X^4 = \kappa_X$ is the kurtosis of $X(t)$, and from Schwartz's inequality: $\kappa_Y \geqslant (3 + \kappa_X - 3) = \kappa_X$. □

Proof of Property 2. For a Gaussian process, $\kappa_X = 0$ in (13). Note that in this case and when $H(t,f)$ is deterministic the Schwartz's inequality becomes an equality, so that $\kappa_Y = 0$ for any non-stationary process driven by a white Gaussian process. □

Proof of Property 3.

$$K_Y(f) = \frac{E\{|H(t,f)|^4\}}{E\{|H(t,f)|^2\}^2}\left(\frac{2\sigma_X^2 + C_{4X}}{\sigma_X^4}\right) - 2 = \gamma_{4H}(f)[2 + \kappa_X] - 2, \quad f \neq 0,$$

which establishes the equality. The inequality follows from the fact that $E\{|H(t,f)|^4\} \geqslant E\{|H(t,f)|^2\}^2$ according to Schwarz's inequality. □

Proof of Property 4. For a Gaussian process, $\kappa_X = 0$ in (25). The inequality holds because, from the Schwarz's inequality, $\gamma_{4H}(f) \geqslant -2$. □

Proof of Property 5. $H(t,f) = H(f) \Rightarrow \gamma_{4H}(f) = 1$. □

Proof of Property 6. For a Gaussian process, $\kappa_X = 0$ in (27). □

Proof of Property 7.

$$K_Y(f_0) = \frac{E\{|m(t)|^4\}}{E\{|m(t)|^2\}^2}\frac{|H(f)|^4}{|H(f)|^4}\frac{S_{4X}}{S_{2X}^2} - 2 = \frac{E\{|m(t)|^4\}}{E\{|m(t)|^2\}^2}\left(\frac{2\sigma_X^4 + C_{4X}}{\sigma_X^4}\right) - 2. \qquad \square$$

Proof of Property 8. Let $H(t,f)\,\mathrm{d}X(f) = \boldsymbol{A}(t)\delta(f - f_0)\,\mathrm{d}f$; then

$$K_Y(f_0) = \frac{E\{|\boldsymbol{A}(t)|^4\}}{E\{|\boldsymbol{A}(t)|^2\}^2} - 2 = \gamma_{4A} - 2. \qquad \square$$

Proof of Property 9.

$$K_Z(f) = \frac{E\{|H(t,f)\,\mathrm{d}X(f) + \mathrm{d}N(f)|^4\}}{E\{|H(t,f)\,\mathrm{d}X(f) + \mathrm{d}N(f)|^2\}^2} - 2$$

$$
\begin{aligned}
&= \frac{S_{4Y}(f) + S_{4N}(f) + 4S_{2Y}(f)S_{2N}(f)}{S_{2Y}^2[1+\rho(f)]^2} - 2 \\
&= \frac{S_{4Y}(f) - 2S_{2Y}^2(f)}{S_{2Y}^2[1+\rho(f)]^2} + \frac{S_{4N}(f) - 2S_{2N}^2(f)}{S_{2Y}^2[1+\rho(f)]^2} \\
&= \frac{K_Y(f)}{[1+\rho(f)]^2} + \frac{\rho(f)^2 K_N(f)}{[1+\rho(f)]^2}, \quad f \neq 0.
\end{aligned}
$$

And, finally according to Property 5, $K_N(f) = K_N$ for a stationary noise. □

Proof of Property 10. For a stationary Gaussian noise, $K_N = 0$ in (25). □

Proof of Proposition 1. Inserting the discrete version of (5) in (36),

$$
Y_w(kP,f) = \sum_{n=-\infty}^{\infty} \int_{-1/2}^{+1/2} H(n,v)\,\mathrm{d}X(v) w(n-kP) \mathrm{e}^{-j2\pi n(f-v)}
$$

Upon condition C1, $H(n,v)$ has slow temporal variations compared to $w(n)$, so that $H(n,v)w(n-kP)$ performs a sampling of $H(n,v)$ at times kP. Hence,

$$
\begin{aligned}
Y_w(kP,f) &\simeq \int_{-1/2}^{+1/2} H(kP,v)\,\mathrm{d}X(v) \sum_{n=-\infty}^{\infty} w(n-kP)\mathrm{e}^{-j2\pi n(f-v)} \\
&= \int_{-1/2}^{+1/2} H(kP,v)\,\mathrm{d}X(v) W(f-v)\mathrm{e}^{-j2\pi kP(f-v)}.
\end{aligned}
$$

Furthermore, upon condition C2 the spectral bandwidth of $w(n)$ is narrow compared to that of $H(n,v)$ (in the frequency variable v). Hence,

$$
Y_w(kP,f) \simeq H(kP,f) \int_{-1/2}^{+1/2} \mathrm{d}X(v) W(f-v)\mathrm{e}^{-j2\pi kP(f-v)}.
$$

The term in the above integral cannot be identified with $\mathrm{d}X(f)$. Indeed, it corresponds to the time sequence $X(n)w(n-kP)$ rather than $X(n)$. However, if the overlap between adjacent analysing windows is small enough, then each $X(n)w(n-kP)$ can be seen as a different realisation of the process $X(n)$, thus leading to a different realisation of the spectral process $\mathrm{d}X(f)$ which we shall denote $\mathrm{d}X(f;k)$. This is the strategy used in the Bartlett/Welch's method of spectral analysis [18]. Therefore,

$$
Y_w(kP,f) \simeq H(kP,f)\,\mathrm{d}X(f;k)
$$

and finally Eq. (37) becomes

$$Y(kP) = w(0)\int_{-1/2}^{+1/2} H(kP,f)\,\mathrm{d}X(f;k). \qquad \square$$

Proof of Proposition 2.

$$\hat{K}_Y(f) = \frac{\hat{S}_{4Y}(f)}{\hat{S}_{2Y}^2(f)} - 2 = \frac{\langle |Y_w(kP,f)|^4\rangle_k}{\langle |Y_w(kP,f)|^2\rangle_k^2} - 2, \quad |f - \mathrm{mod}(1/2)| > 1/N_w.$$

From Slutsky's theorem on probability limits, the infinite time-average $\langle\cdots\rangle_k$ leads to

$$E\{\hat{K}_Y(f)\} = \frac{E\{|Y_w(kP,f)|^4\}}{E\{|Y_w(kP,f)|^4\}^2} - 2, \quad |f - \mathrm{mod}(1/2)| > 1/N_w.$$

Under conditions C1 and C2, it has been established that

$$Y_w(kP,f) \simeq H(kP,f)\int \mathrm{d}X(v)W(f-v)\mathrm{e}^{j2\pi kP(f-v)}.$$

Hence,

$$E\{|Y_w(kP,f)|^2\} \simeq S_{2H}(f)\iint E\{\mathrm{d}X(v_1\,\mathrm{d}X(v_2)^*\}W(f-v_1)W(f-v_2)^*\mathrm{e}^{j2\pi kP(v_2-v_1)}$$

with $S_{2nH}(f) \triangleq E\{|H(t,f)|^{2n}\}$. Since $E\{\mathrm{d}X(v_1)\,\mathrm{d}X(v_2)^*\} = \mathrm{d}v_1\,\mathrm{d}v_2\sigma_X^2\delta(v_1-v_2)$ for a stationary white noise,

$$E\{|Y_w(kP,f)|^2\} \simeq S_{2H}(f)\sigma_X^2\int |W(f-v)|^2\,\mathrm{d}v = S_{2H}(f)\sigma_X^2 E_w$$

with $E_w \triangleq \sum_n |w(n)|^2$. Similarly,

$$\begin{aligned} E\{|Y_w(kP,f)|^4\} \simeq S_{4H}(f)\iiint & E\{\mathrm{d}X(v_1)\,\mathrm{d}X(v_2)^*\,\mathrm{d}X(v_3)\,\mathrm{d}X(v_4)^*\} \\ & \ldots W(f-v_1)W(f-v_2)^*W(f-v_3)W(f-v_4)^*\mathrm{e}^{j2\pi kP(v_2-v_1+v_3-v_4)} \end{aligned}$$

and since $E\{\mathrm{d}X(v_1)\,\mathrm{d}X(v_2)^*\,\mathrm{d}X(v_3)\,\mathrm{d}X(v_4)^*\} = \mathrm{d}v_1\,\mathrm{d}v_2\,\mathrm{d}v_3\,\mathrm{d}v_4\,[\sigma_X^4\delta(v_1-v_2)\delta(v_3-v_4) + \sigma_X^4\delta(v_1+v_3)\delta(v_2+v_4) + \sigma_X^4\delta(v_1-v_3)\delta(v_2-v_4) + C_{4X}\delta(v_1-v_2+v_3-v_4)]$ for a stationary white noise of order $p \geqslant 4$,

$$\begin{aligned} E\{|Y_w(kP,f)|^4\} \simeq S_{4H}(f)\Bigg[& 2\sigma_X^4\left(\int |W(f-v)|^2\,\mathrm{d}v\right)^2 + \sigma_X^4\left|\int W(f+v)W(f-v)\,\mathrm{d}v^2\right| \\ & + \cdots\iiint C_{4X}W(f-v_1)W(f-v_2)^*W(f-v_3) \\ & \times W(f-v_1+v_2-v_3)^*\,\mathrm{d}v_1\,\mathrm{d}v_2\,\mathrm{d}v_3\Bigg] \\ = S_{4H}(f)[2\sigma_X^4E_w^2 & + 0 + C_{4X}E_{w^2}], \quad |f - \mathrm{mod}(1/2)| > 1/N_w \end{aligned}$$

with $E_{w^2} \triangleq \sum_n |w(n)|^4$. Finally,

$$E\{\hat{K}_Y(f)\} = \frac{S_{4H}(f)}{S_{2H}^2(f)}\left(2 + \frac{C_{4X}E_{w^2}}{\sigma_X^4 E_w^2}\right) - 2, \quad |f - \mathrm{mod}(1/2)| > 1/N_w.$$

The rest of the proof follows by using Property 3. □

Proof of Proposition 3. Using perturbation calculus on large sample results,

$$\frac{\mathrm{Var}\{\hat{K}_Y(f)\}}{K_Y(f)+2} \simeq \frac{\mathrm{Var}\{\hat{S}_{4Y}(f)\}}{S_{4Y}^2(f)} + 4\,\frac{\mathrm{Var}\{\hat{S}_{2Y}(f)\}}{S_{2Y}^2(f)} - 4\,\frac{\mathrm{Cov}\{\hat{S}_{4Y}(f), \hat{S}_{2Y}(f)\}}{S_{4Y}(f)S_{2Y}(f)}.$$

Since the K blocks of data used in the STFT have been assumed independent (no overlap), it can be shown that:

$$\frac{\mathrm{Var}\{\hat{K}_Y(f)\}}{K_Y(f)+2} \simeq \frac{E\{|Y_w(kP,f)|^8\}}{K \cdot S_{4Y}^2(f)} + 4\,\frac{E\{|Y_w(kP,f)|^4\}}{K \cdot S_{2Y}^2(f)} - 4\,\frac{E\{|Y_w(kP,f)|^4 |Y_w(kP,f)|^2\}}{K \cdot S_{4Y}(f)S_{2Y}(f)}.$$

From the assumption of Gaussianity, and proceeding as in the proof of Proposition 1, $E\{|Y_w(kP,f)|^8\} = M_{8H}(f) \cdot 24\sigma_X^8 E_w^4$, $E\{|Y_w(kP,f)|^4\} = M_{4H}(f) \cdot 2\sigma_X^4 E_w^2$, and $E\{|Y_w(kP,f)|^4 | Y_w(kP,f)|^2\} = M_{6H}(f) \cdot 6\sigma_X^6 E_w^3$. The rest of the proof follows by inserting $S_{4Y}(f) = S_{4H}(f) \cdot 2\sigma_X^4 E_w^2$ and $S_{2Y}(f) = S_{2H}(f) \cdot \sigma_X^2 E_w$. □

Proof of Property 11. Let us first suppose that $pN_w \ll 1$ so that there is a negligible probability that two impulses are separated by less than N_w samples. Hence,

$$\hat{K}_y(f) = \frac{\langle |Y_w(kP,f)|^4 \rangle_k}{\langle |Y_w(kP,f)|^2 \rangle_k^2} - 2 \underset{k}{\rightarrow} \frac{E\{|Y_w(kP,f)|^4/I\}P(I) + E\{|Y_w(kP,f)|^4/\bar{I}\}P(\bar{I})}{(E\{|Y_w(kP,f)|^2/I\}P(I) + E\{|Y_w(kP,f)|^2/\bar{I}\}P(\bar{I}))^2} - 2,$$

where $P(I)$ is the probability that one impulse occurs in the interval covered by $w(n-kP)$. $P(I)$ may be found as the ratio of the number of impulses to the number of windows times the average number of windows shared by one impulse, i.e.

$$\frac{p}{1/P} \times \frac{N_w}{P} = pN_w. \tag{53}$$

Therefore,

$$\hat{K}_y(f) \underset{k}{\rightarrow} \frac{E\{|Y_w(kP,f)|^4/I\}pN_w + 0}{(E\{|Y_w(kP,f)|^2/I\}pN_w + 0)^2} = \frac{E\{|Y_w(kP,f)|^4/I\}}{E\{|Y_w(kP,f)|^2/I\}^2}\,\frac{1}{pN_w} - 2.$$

The rest of the proof follows by noting that $E\{|Y_w(kP,f)|^{2n}/I\} = \frac{1}{N_w}\sum_k |w(k)|^{2n}$. □

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